



MASTER GEOMATIC AND SPATIAL MODELISATION

Course: GIS AND SPATIAL ANALYSIS

GIS AND SPATIAL ANALYSIS: WILDFIRES, VEGETATION, URBANIZATION, AND AGRICULTURE

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GENERAL INTRODUCTION

Spatial analysis is now a fundamental tool for understanding territorial and environmental dynamics, particularly in contexts subject to strong natural and human-induced pressures. Through the use of geographic data and geographic information systems (GIS), it is possible to combine information of various kinds to identify spatial structures, trends, and areas of concern. This project is part of this approach, employing spatial analysis methods applied to major environmental and territorial themes, namely vegetation, wildfires, agriculture, and urban space.

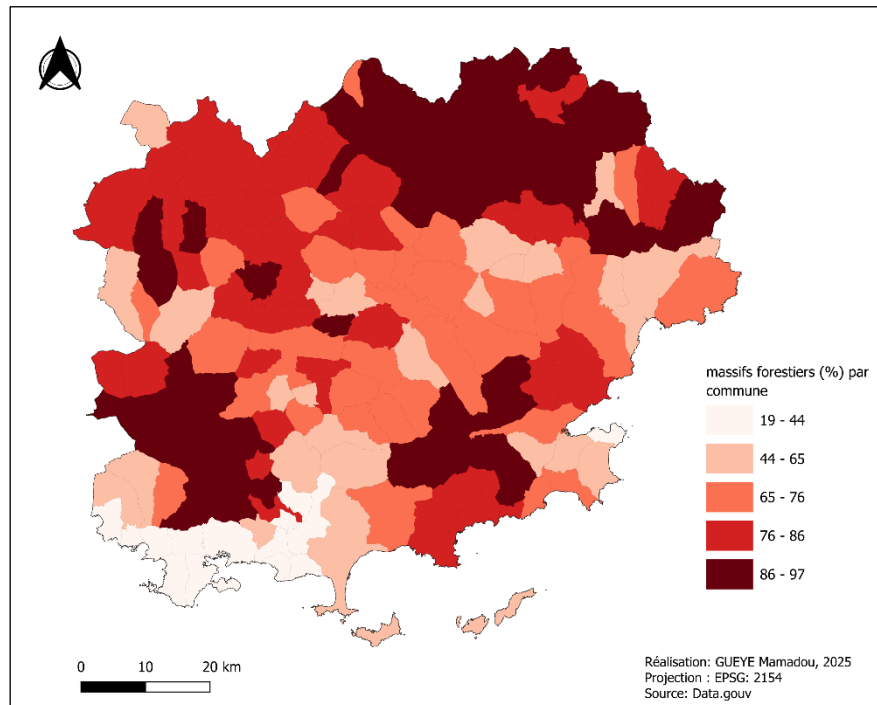
The study is conducted at several spatial scales to better understand the diversity of the phenomena analyzed. The themes of vegetation, wildfires, and agriculture are addressed at the departmental level, in the Var region, characterized by a high vulnerability to natural hazards and contrasting land use. The urban analysis, on the other hand, was conducted at a finer scale within the city of Nice, allowing for a detailed understanding of urban dynamics. To achieve this, various geographic databases were used, including the BD TOPO, Corine Land Cover, the Registre Parcellaire Graphique (Graphic Parcel Register), and the BD Parcellaire. Processing and analysis were carried out using QGIS, office software (Excel), and Python scripts developed on Google Colab, thus combining cartographic, statistical, and automated analyses.

I. VEGETATION ANALYSIS IN THE VAR DEPARTMENT

Vegetation analysis is a central element of this project, as it allows us to understand the state and distribution of vegetation cover across the Var department. This Mediterranean region is characterized by a great diversity of plant formations, strongly influenced by climatic conditions, topography, and human activities. Through the use of land cover data and reference geographic databases, this section aims to analyze the spatial distribution of vegetation, identify the main types of vegetation cover, and highlight the associated territorial dynamics.

1. CARTOGRAPHIC ANALYSIS

Distribution of forest areas in the VAR department

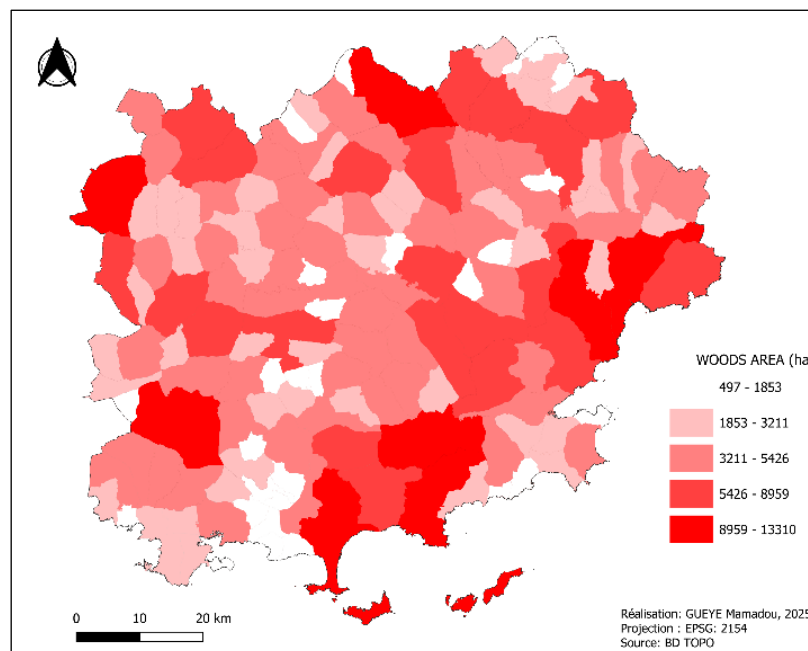


Map 1: Distribution of forest areas (by municipality) in the VAR

The map highlights a contrasting distribution of forest areas within the municipalities of the Var department. The highest percentages of forest cover (classes 86–97%) are concentrated mainly in the northern and northeastern parts of the department, where many municipalities have very dominant forest cover. These areas appear continuous, forming vast, relatively unfragmented forest tracts. Conversely, municipalities located on the coast and in the southwest of the Var show significantly lower percentages of forest cover, primarily in the 19–44% and 44–65% classes. These areas exhibit a

marked discontinuity in forest vegetation, with some municipalities having very little woodland. The central areas of the department present an intermediate situation, with a mosaic of municipalities belonging to the 65–76% and 76–86% classes, reflecting significant but less dominant forest cover than in the north.

This spatial organization can be explained by several territorial factors. The municipalities with extensive forest cover are mostly located in less urbanized, predominantly natural areas, where physical constraints and distance from major urban centers limit other forms of land use. Conversely, the low percentages of forest cover observed along the coast and in certain southern areas are found in more densely developed territories, characterized by urbanization, infrastructure, and human activities.



Map 2: Representation of forest areas by municipality (VAR department)

This map represents the total area of wooded land per municipality, expressed in hectares, and highlights significant spatial heterogeneity within the Var department. The highest values (classes 8,959–13,310 ha) are concentrated in several large municipalities, primarily located in the northern, eastern, and southeastern parts of the department, as well as in certain southern areas.

Compared with the map representing forest areas as a percentage, an overall consistency is observed in the spatial distribution of the most forested municipalities. The municipalities with the largest wooded areas are also, for the most part, among

those with high rates of forest cover. This correlation indicates that municipalities characterized by vast forest areas are not only marked by the absolute importance of forests, but also by their strong relative dominance within the municipal land use. The two representations thus complement each other and confirm the predominance of forest areas in these territories.

2. STATISTICAL ANALYSIS

Figure 1: Dominant vegetation type in the Var (surface)



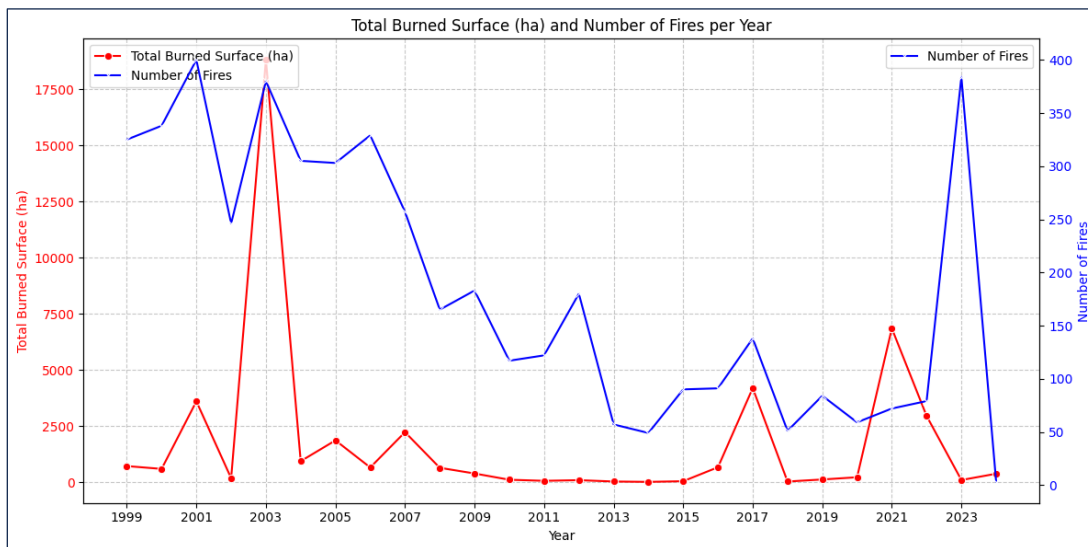
The graph highlights a marked differentiation between the main types of forest vegetation considered in the analysis. Closed mixed forests appear as the most represented category, followed by open woodlands, reflecting a strong presence of diverse forest formations. Closed coniferous forests and closed deciduous forests show intermediate numbers, indicating a more moderate distribution of these types of vegetation cover. Finally, wooded heaths occupy a significant area but remain less dominant, illustrating the structural diversity of the forest composition across the studied territory.

II. ANALYSIS OF THE VAR WILDFIRE

1. STATISTICAL ANALYSIS

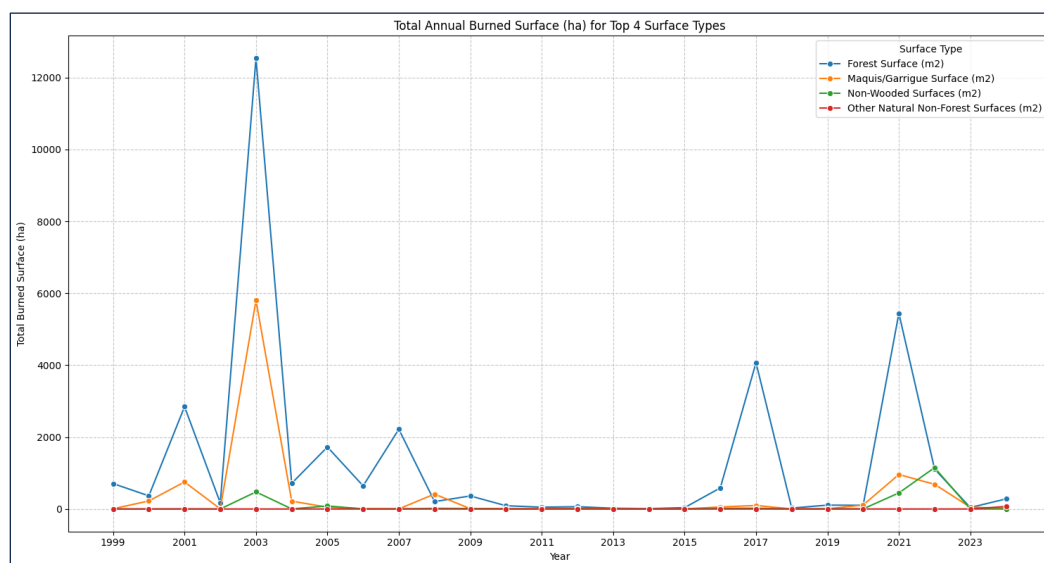
Wildfires represent a major environmental challenge in the Var department, due to the combination of Mediterranean climatic conditions, dense vegetation cover, and increasing human pressures. This part of the project aims to analyze the spatial distribution and evolution of wildfires using the BDIFF database, covering the period from 1999 to 2025. The analysis of this data allows us to identify the spatio-temporal dynamics of fires at the departmental level, before focusing on a case study located in the municipality of Bormes-les-Mimosas, in order to concretely illustrate the mechanisms and challenges associated with wildfires.

Figure 2: Curves of burned areas as a function of the number of fires per year.



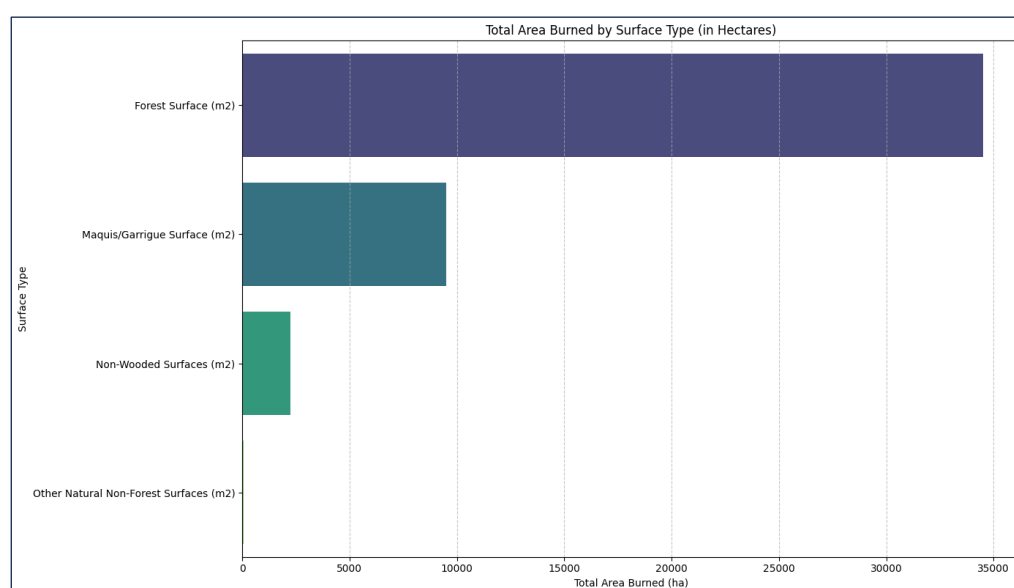
This graph correlates the total annual burned area with the number of recorded fires. It shows that years characterized by very large burned areas, such as 2003, 2017, or 2021, do not necessarily correspond to the years with the highest number of fires. Conversely, some years show a high number of fires for relatively small burned areas. This disparity highlights that the spatial impact of fires depends more on a few major events than solely on the frequency of fires. The period after 2010 is also distinguished by an overall decrease in burned areas, despite persistent fluctuations in the number of fires.

Figure 3: Total surface area of the most consumed species types



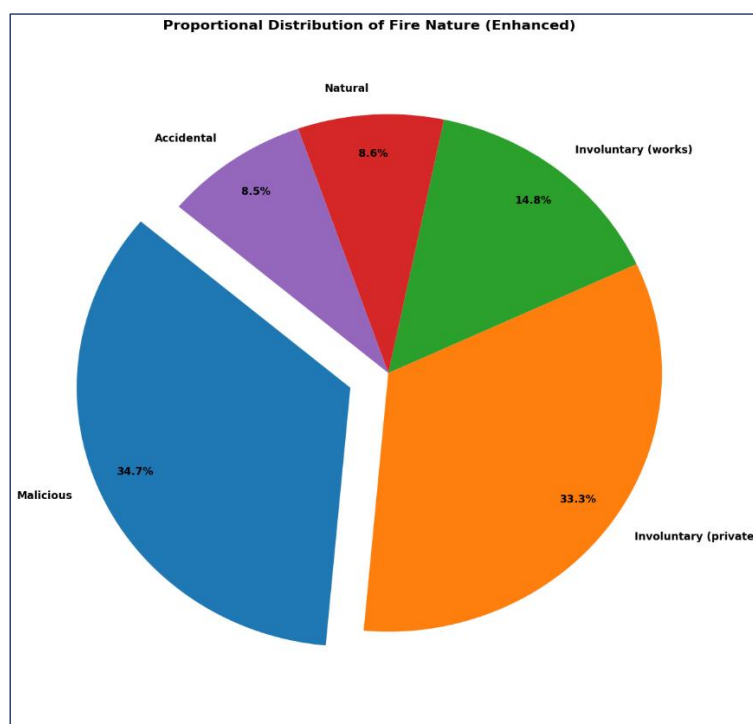
However, this graph highlights significant year-to-year variability in burned areas according to vegetation cover type between 1999 and 2025. Forested areas consistently appear as the most affected, with marked peaks in certain years, notably 2003, 2017, and 2021, when burned areas reached values significantly higher than other categories. Maquis and garrigue vegetation formations constitute the second most affected type of cover, generally following the same temporal dynamics as forests, but with more limited burned areas. Non-forested areas and other natural areas remain marginal throughout the period, with low and sporadic values.

Figure 4 : Nombre total des surface brulées en ha



This figure provides a very direct illustration of the impact of wildfires according to land cover type over the entire study period. It is immediately apparent that forested areas account for the vast majority of burned zones, with a cumulative area significantly greater than that of other categories, exceeding 30,000 hectares between 1999 and 2025. Maquis and garrigue vegetation formations are less affected, with significant but considerably smaller areas impacted than forests. Conversely, non-wooded areas and other non-forested natural spaces appear to be very little affected by fires. This distribution highlights the central role of dense vegetation formations in fire propagation and underscores that fires in the Var department primarily impact forested and semi-natural areas, which constitute the core of the fire risk at the departmental level.

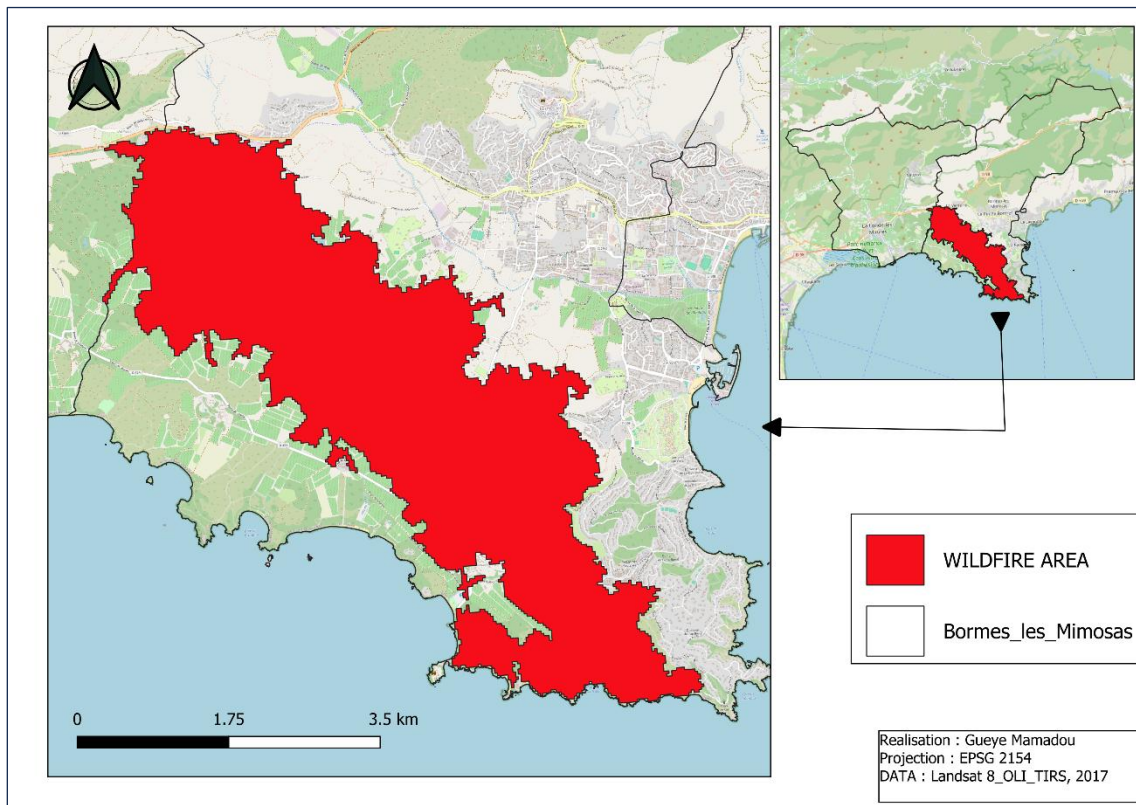
Figure 5: Proportion of major causes of fires in the Var



This figure clearly shows that human activities are fundamentally the leading cause of wildfires in the Var department. The majority of fire sources confirm this. It is observed that more than 34% of fire ignitions are due to malicious acts, and that unintentional fires (individuals) rank second as a cause of fires in the Var between 1999 and 2025. The other types of fires have lower proportions but are still not negligible.

2. CASE STUDY OF A FIRE: BORMES-LES-MIMOSAS

The municipalities of Bormes-les-Mimosas and La Londe-les-Maures are located in the Var department (83), in the Provence-Alpes-Côte d'Azur (PACA) region, in southeastern France. They are situated on the Mediterranean coast, in close proximity to the city of Hyères and not far from the Gulf of Saint-Tropez. Bormes-les-Mimosas borders the municipalities of La Londe-les-Maures, Le Lavandou, and Collobrières.



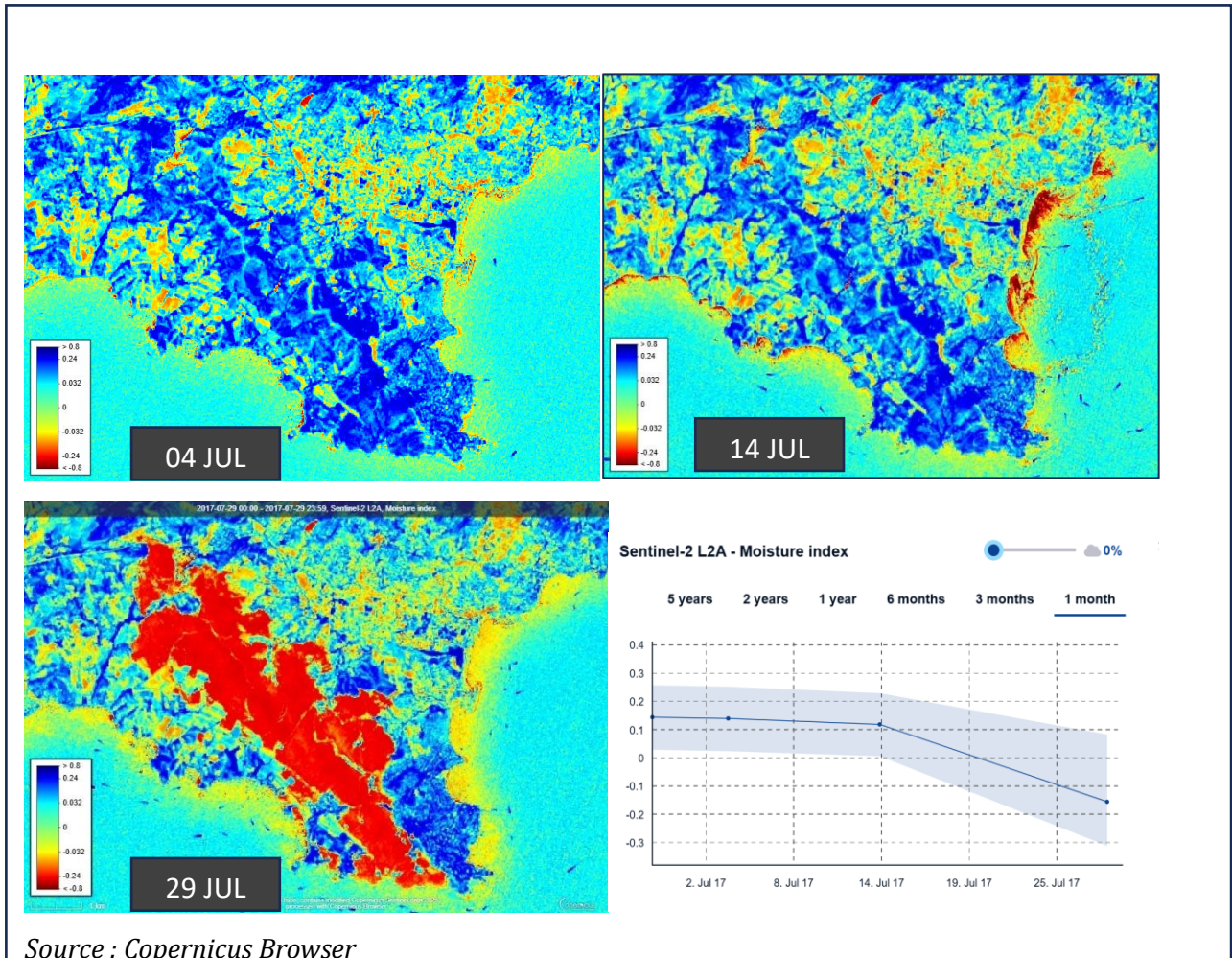
Map 3: Representation of the Bormes-les-Mimosas fire

The summer of 2017 was marked by exceptional wildfire activity across the entire Mediterranean region. In the Var, a department particularly exposed to wildfires, more than 4,000 hectares were ravaged by flames.

It was in this context that the Bormes-les-Mimosas fire broke out on the night of July 24 to 25, 2017. This fire was part of a series of particularly violent blazes that affected southeastern France during this period, fueled by extreme weather conditions: intense heat, prolonged drought, and strong winds, particularly the mistral, greatly contributed to the rapid spread of the flames. The fire spread over a vast area, mainly affecting the municipalities of Bormes-les-Mimosas and La Londe-les-Maures. However, Bormes-les-

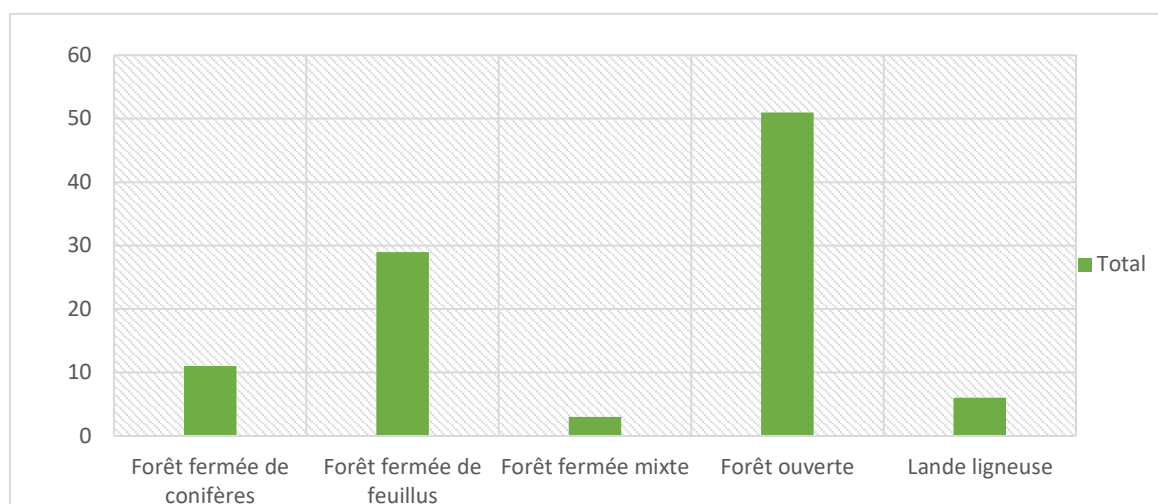
Mimosas was the hardest hit, which explains why this event is referred to as the Bormes-les-Mimosas fire.

Figure 6: Images of the burned area before and after the fire



By analyzing these images extracted from the Copernicus Browser, we can observe the condition of the vegetation in the burned area before the fires using the moisture index. If clouds are completely removed from the images, a consistent vegetation moisture level can be observed, ranging between 0.1 and 0.2 at the beginning of the month, then dropping sharply after July 15 to reach -0.17 on the 29th. This situation in turn increases the risk of fire spread after ignition, as the drier the vegetation, the more vulnerable it becomes to wildfires.

Figure 7: Share of each vegetation type in the burned area



The graph shows that open forest largely dominates the burned area, accounting for more than half of the affected surface (around 51%). This means that this type of vegetation, often less dense but highly flammable, was the most impacted by the fire. This type of environment generally consists of low vegetation, shrubs, and sparse woods, which promote rapid fire spread.

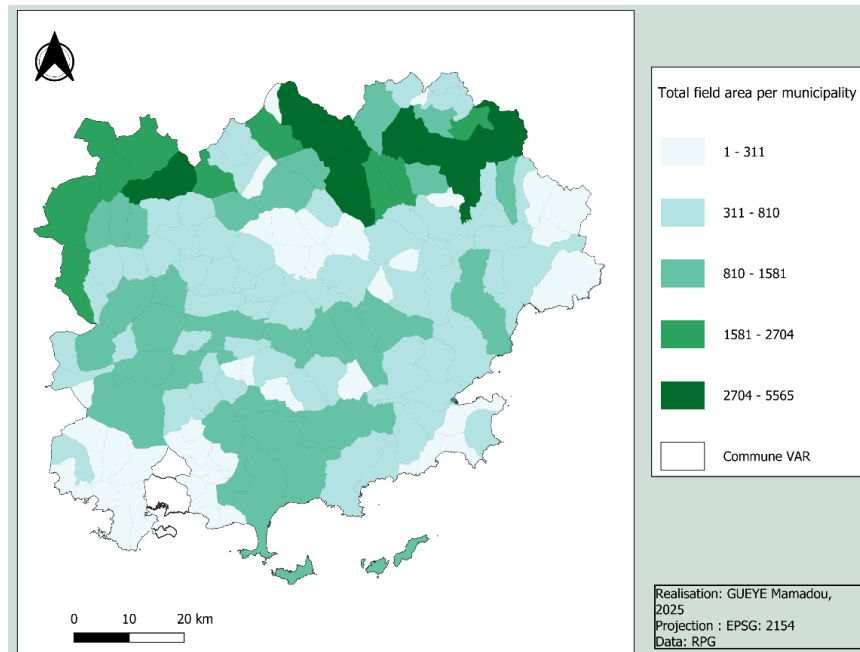
In second place, closed deciduous forest represents nearly 29% of the burned vegetation. Deciduous forests, although slightly more resistant to drought than conifers, can also ignite quickly in summer, especially if the undergrowth is dense and dry. Closed coniferous forests (about 11%) are also well represented. Conifers, such as pines or firs, produce highly flammable resins, explaining their strong vulnerability to fires. Woody heathlands (~7%) and mixed forests (~3%) are less represented, but their presence indicates some diversity in the affected vegetation types.

This distribution shows that the fire affected a wide variety of forest environments, with a predominance of open vegetation. It highlights both the vulnerability of these Mediterranean landscapes to fires and the importance of vegetation management to reduce risks. The significant presence of open forests also indicates that the landscape structure facilitated rapid fire spread, making firefighting operations more difficult.

III. AGRICULTURAL ANALYSIS

In this section, the goal is to characterize the spatial organization of agricultural activities within the Var department and to better understand their role in land use. Using data from the Graphical Parcel Register and other reference databases, this section allows for the identification of the main crop types and their geographical distribution. The approach

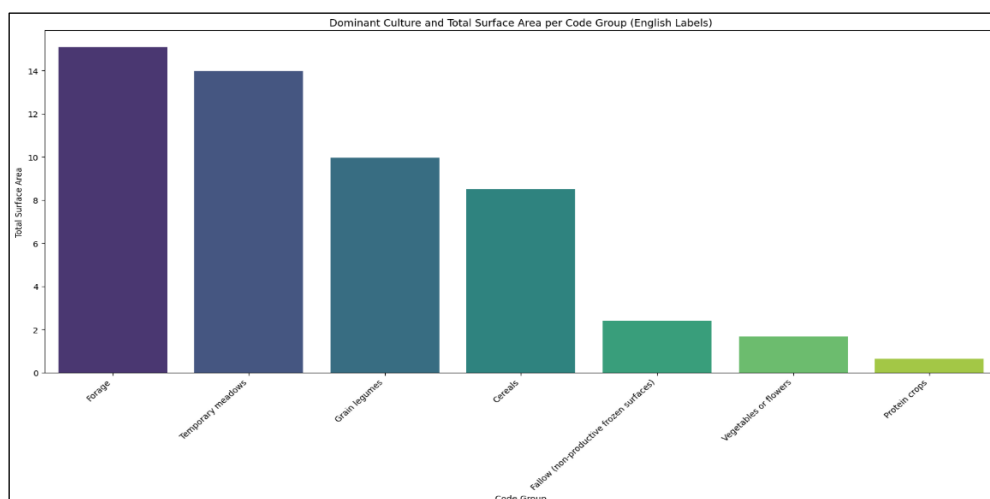
adopted relies on GIS processing and statistical analyses to highlight territorial contrasts and the patterns of agricultural land placement.



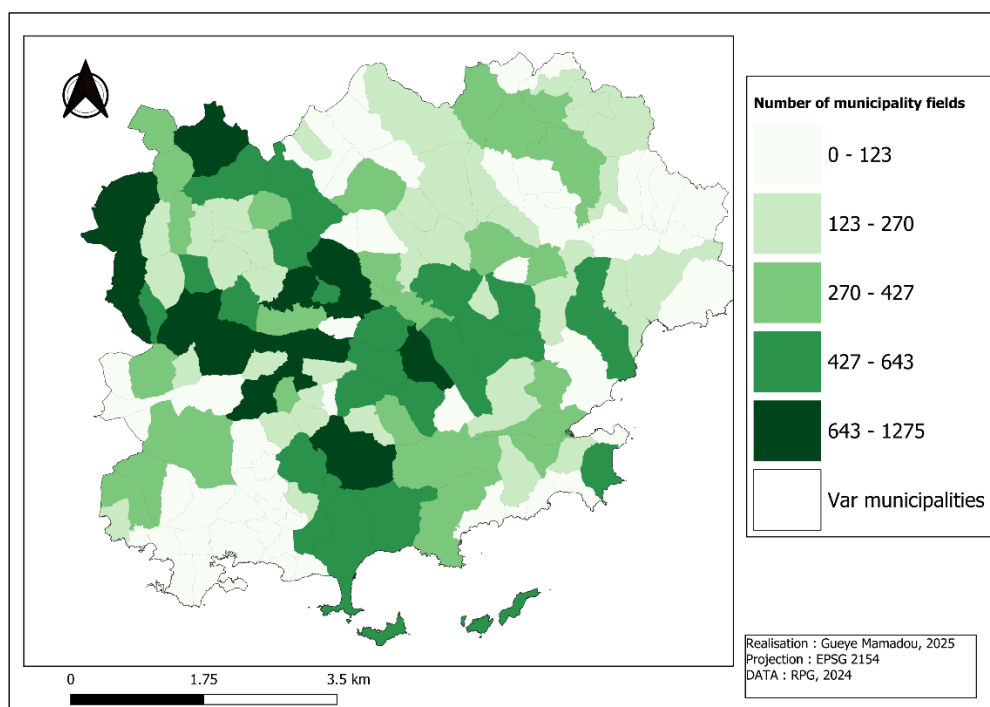
Map 4: Distribution of agricultural areas by municipality in the Var department

This map illustrates the distribution of total agricultural areas by municipality in the Var department and highlights a contrasting spatial organization. The highest values (classes 2,704–5,565 ha) are mainly concentrated in the northern part of the department, where several municipalities show a strong dominance of agricultural land. These areas form relatively continuous zones, reflecting a marked agricultural vocation. In contrast, municipalities with the lowest agricultural areas (1–311 ha) are mostly located along the coast and in certain southern zones, where land use is more oriented toward other purposes. The central areas of the department show intermediate values, indicating a more balanced coexistence between agricultural activities and other forms of land use.

Figure 8: Dominant crop types in the department by area



Regarding the dominant crop types in the Var department based on their total area, permanent crops constitute the most represented category, reflecting the significant role of long-term agricultural systems in the studied territory. They are followed by temporary crops, which also occupy a substantial portion of agricultural land. Pastures and cereals cover intermediate areas, indicating a more moderate presence of these crop types. In contrast, mixed farms, vegetable and horticultural crops, as well as protein crops occupy much smaller areas, highlighting their more marginal role in the department's agricultural structure.



Map 5: Representation of the number of agricultural parcels by municipality (Var)

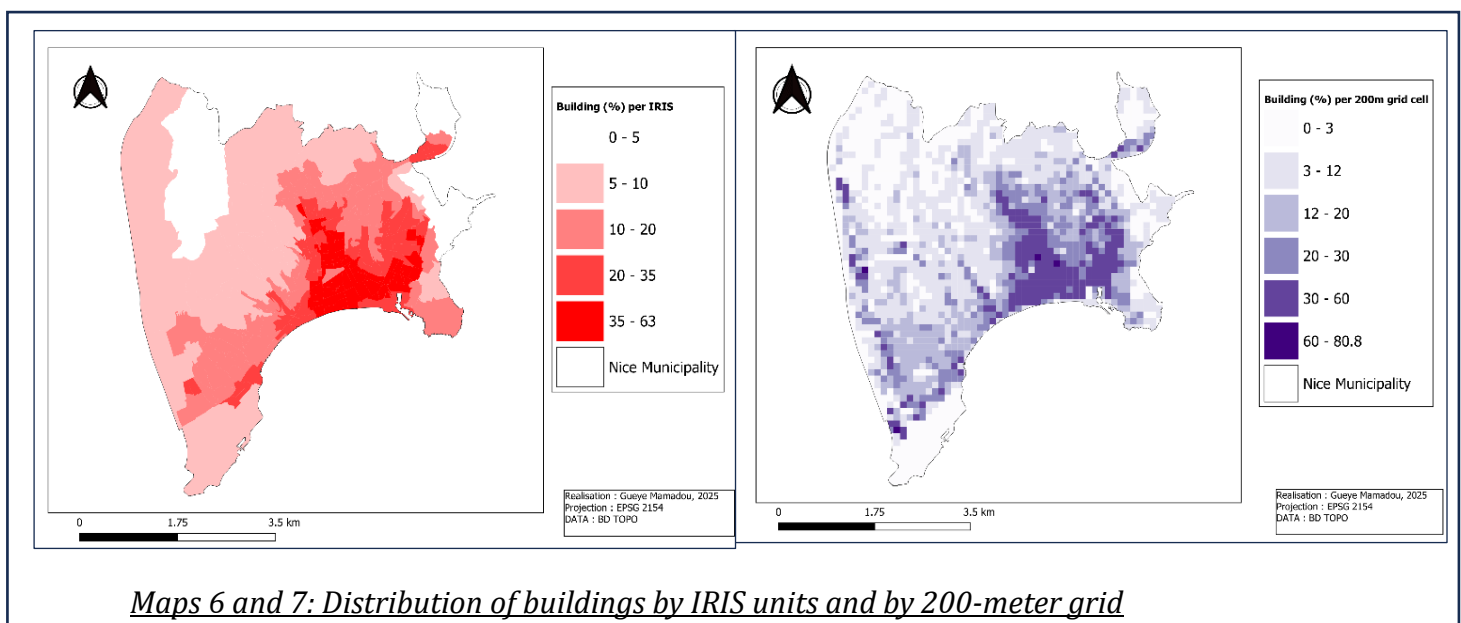
This map represents the number of agricultural parcels per municipality in the Var department and highlights a heterogeneous spatial distribution. The highest values (643–1,275 parcels) are mainly concentrated in the central and western areas of the department, where several municipalities show significant fragmentation of agricultural land. These areas contrast with municipalities showing the lowest values (0–123 parcels), mostly located along the coast and in certain southern parts, reflecting a more limited or less fragmented agricultural presence.

Comparing this map with the previous one, it is evident that municipalities with large agricultural areas do not necessarily correspond to those with the highest number of parcels, highlighting differences in parcel structure. Some municipalities stand out for having large agricultural areas with relatively few parcels, while others combine smaller areas with a high number of parcels.

IV. URBAN ANALYSIS

1. BUILDING DENSITY

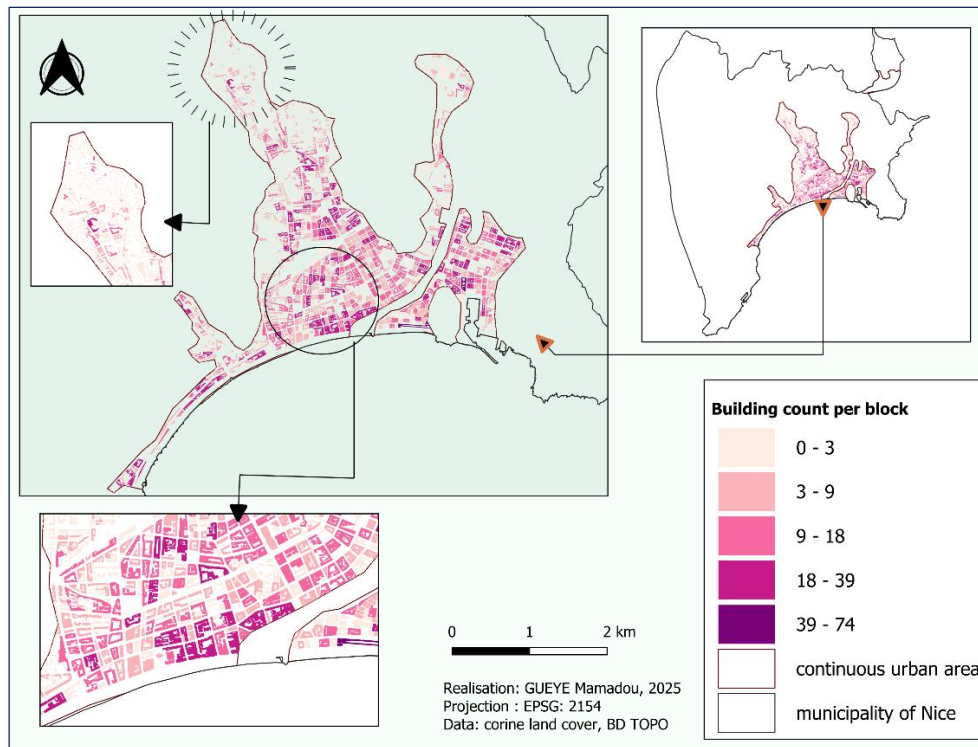
The urban section of the project aims to analyze the spatial organization and characteristics of the urban fabric at a finer scale, focusing on the municipality of Nice. For most of the analyses, two types of scales will be used: that of the IRIS units and a 200-meter grid. This change of scale allows for a more precise understanding of urban dynamics, land use patterns, and planning logic. Using reference geographic data and GIS processing, this analysis highlights the structuring of urban space and internal contrasts within the municipality, providing a detailed view of urban challenges.



The first map represents the share of built-up areas per IRIS unit in the municipality of Nice. It highlights a strong concentration of buildings in the IRIS units located in the city center and along the coastline, where rates reach the highest classes (35 to 63%). These areas form a spatially continuous zone, reflecting dense and structured urbanization. In contrast, the peripheral IRIS units, particularly in the west and northern parts of the municipality, show lower building rates, often between 5 and 20%, or even below 5%.

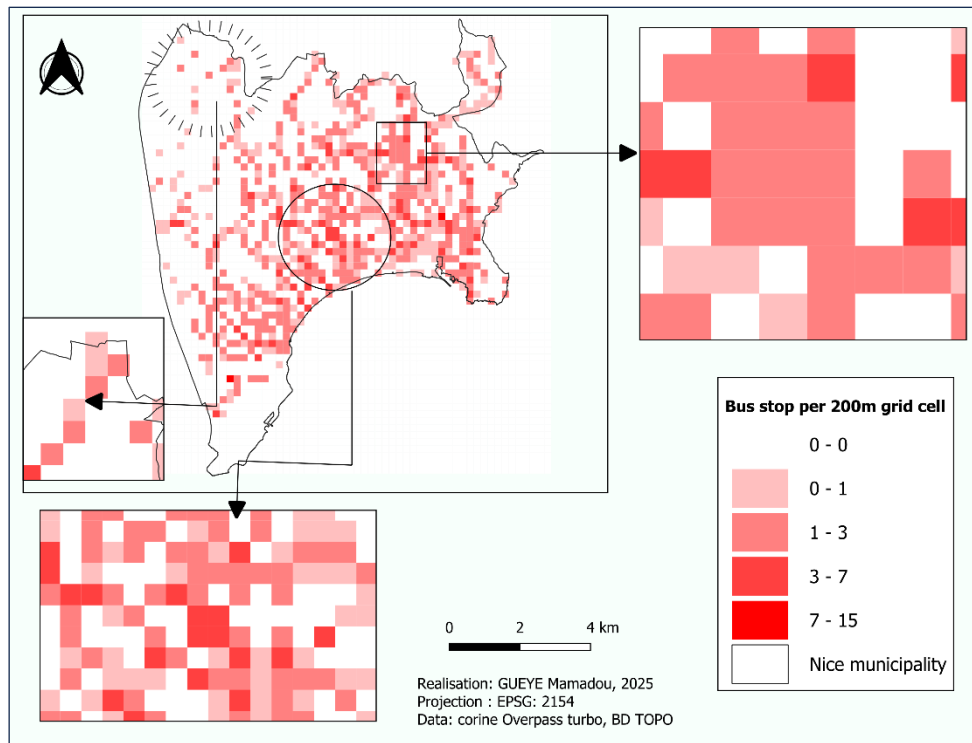
The second map provides a finer-scale view of building coverage, calculated using a regular 200-meter grid. It reveals marked intra-urban heterogeneity, with strong contrasts even within individual IRIS units. The highest values (60 to 80.8%) appear as localized clusters, mainly concentrated in the city center and certain coastal sectors. Conversely, sparsely built areas appear as large light-colored zones, particularly in peripheral areas and elevated terrain. This representation thus highlights the spatial discontinuity of built-up areas at a fine scale, which is not visible at a more aggregated scale.

Comparing the two maps shows that the IRIS-based analysis highlights the major structures of urbanization, while the 200-meter grid reveals strong local heterogeneity in building density. The most urbanized areas are consistent in both representations but appear more fragmented at the finer scale. This comparison underscores the importance of scale choice in the spatial analysis of the urban fabric.



Map 8: Distribution of the number of buildings per block in the continuous urban fabric of Nice

This map highlights the spatial organization of the continuous urban fabric of the municipality of Nice, with its boundary extracted from the Corine Land Cover database. The representation of the number of buildings per block reveals a high building density in the urban core (39 to 74 buildings per block), particularly in the central and coastal areas, where the highest classes are continuously concentrated. In contrast, the edges of the continuous urban fabric show lower densities and a looser building arrangement, reflecting a gradual transition to less compact urban forms. Insets and map zooms emphasize internal contrasts within the urban fabric, highlighting marked heterogeneity even within the continuous urban area. In summary, building density decreases progressively as one moves away from the city center.

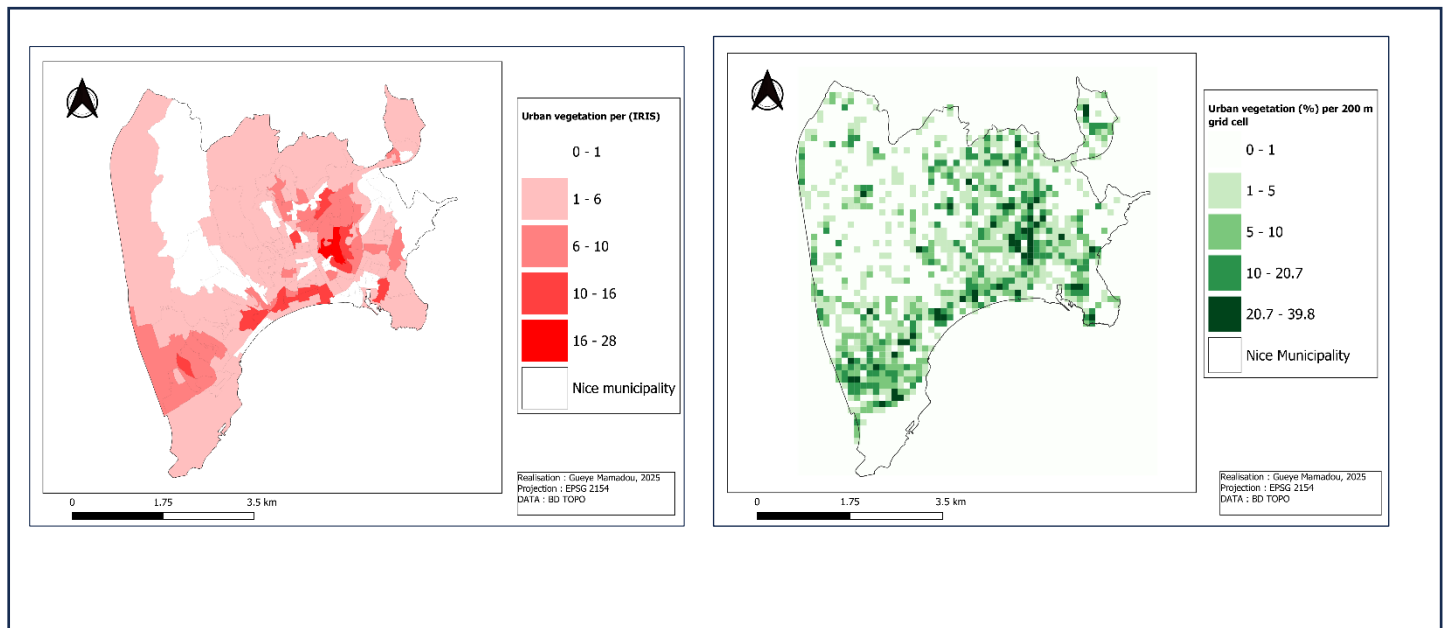


Map 9: Distribution of bus stops in the municipality of Nice using a 200-meter grid

This map represents the density of bus stops within the municipality of Nice, calculated using a regular 200-meter grid. It highlights a strong spatial heterogeneity in public transport provision. The highest values (classes 7–15 stops per grid cell) are mainly concentrated in the central and coastal parts of the municipality, corresponding to the most urbanized areas structured by the main road network. These zones form continuous high-density cores, reflecting high accessibility to public transport. In contrast, grid cells with few or no stops (classes 0–0 and 0–1) are mostly located in peripheral areas, particularly in the north and in some marginal zones of the dense urban fabric, where coverage is more discontinuous.

Multi-scale analysis illustrated by the zooms confirms this spatial organization. At a larger scale, bus stop density is structured around main transport axes, with a gradual decrease toward areas farther from the center. The contrasts observed between adjacent grid cells also reveal strong local variability in accessibility, highlighting micro-imbalances in the distribution of transport infrastructure. This map thus illustrates a hierarchical structuring of the bus network, closely linked to urban intensity and the urban morphology of the municipality of Nice.

2. Urban vegetation analysis



Maps 10 and 11: Share of woodland and hedges at the IRIS scale and by 200-meter grid

These two maps represent the distribution of urban vegetation in the municipality of Nice at two different scales. The first map concerns the IRIS scale, showing a non-uniform distribution of urban vegetation (woodland and hedges). A higher concentration is visible in part of the central-west area, with values between 16 and 28%. However, IRIS units with values between 1 and 6% remain dominant overall.

The map on the right uses a regular 200-meter grid, allowing a more detailed view of vegetation distribution. It shows that some small areas have a high presence of vegetation, even if they are located in generally sparsely vegetated zones. Additionally, in areas where the urban fabric becomes discontinuous, vegetation density is much higher. The IRIS division provides a useful overall view but can hide variations within the same zone. In contrast, the grid representation offers a more precise picture, revealing small highly vegetated spaces that are sometimes invisible at the IRIS scale.

This comparison shows that urban vegetation in Nice is unevenly distributed: it is more abundant in peripheral neighborhoods and less present in the city center. This can have significant implications for thermal comfort, quality of life, and environmental equity for residents.

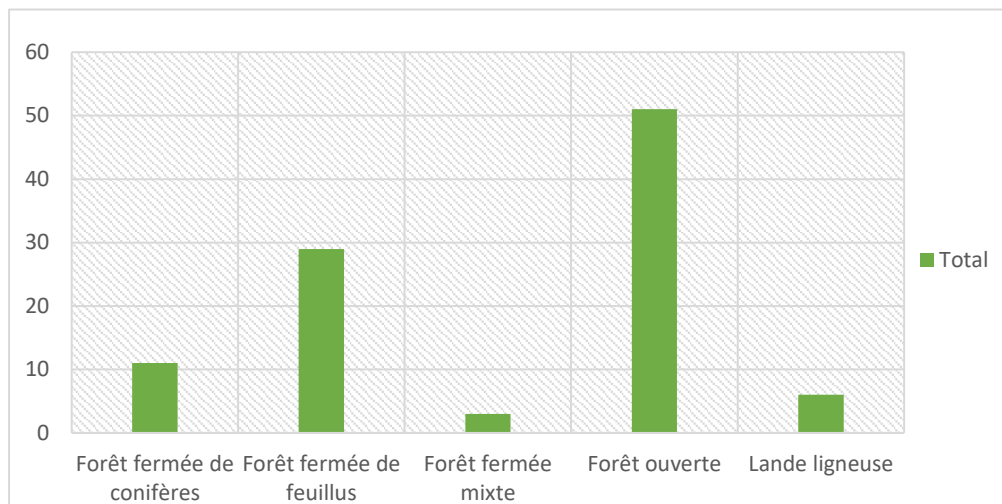


Figure 9: Share of each vegetation type in the burned area

The graph shows that open forest largely dominates the burned area, accounting for more than half of the affected surface (around 51%). This means that this type of vegetation, often less dense but highly flammable, was the most impacted by the fire. This environment generally consists of low vegetation, shrubs, and sparse woods, which promote rapid fire spread.

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This distribution shows that the fire affected a wide variety of forest environments, with a predominance of open vegetation. It highlights both the vulnerability of these Mediterranean landscapes to fires and the importance of managing vegetation cover to reduce risks. The significant presence of open forests also indicates that the landscape structure facilitated rapid fire spread, making firefighting operations more difficult.

GENERAL CONCLUSION

This spatial analysis project made it possible to explore environmental, agricultural, and urban dynamics in a cross-cutting manner through a multi-thematic and multi-scale approach. At the departmental scale, the study of vegetation highlighted a strong dominance of forest massifs in the inland areas of the Var, both in proportion and in surface area, contrasting with more artificialized coastal spaces. The analysis of forest fires, based on historical data from the BDIF, emphasized the vulnerability of these heavily wooded territories to fire risk, while also illustrating, through the case of Bormes-les-Mimosas, the value of a localized approach to better understand the spatial dynamics of fires. In addition, the agricultural analysis revealed a contrasted organization of cultivated areas, characterized both by differences in agricultural surface areas and by varied parcel structures, highlighting the diversity of agricultural systems within the department.

The change of scale applied in the urban section, focused on the municipality of Nice, provided complementary insight into territorial dynamics at a finer scale. The analysis of the distribution of bus stops revealed a strong concentration of transport services in central and densely urbanized areas, contrasted with more diffuse coverage in peripheral spaces. Overall, the results highlight the value of spatial analysis and GIS tools for combining diverse types of data, identifying territorial contrasts, and revealing spatial organization patterns. This integrated approach constitutes essential support for understanding planning, environmental management, and territorial planning issues in contexts subject to strong natural and human pressures.